

PORTFOLIO

Arizqi Ramadhana

introduction

Hi, I am **ARRIZQI PRAMADHANA**

A dedicated Multimedia Engineer (GPA 3.81/4.00) with deep expertise in Augmented Reality (AR) development and interactive multimedia systems. I specialize in bridging creative design with technical engineering, utilizing Unity (C#) and 3D Modeling (Blender) to build immersive digital solutions. From developing AR wayfinding applications to R&D in AI-based NPC behaviors, I am committed to delivering high-impact, user-centric technology that transforms complex ideas into functional reality.

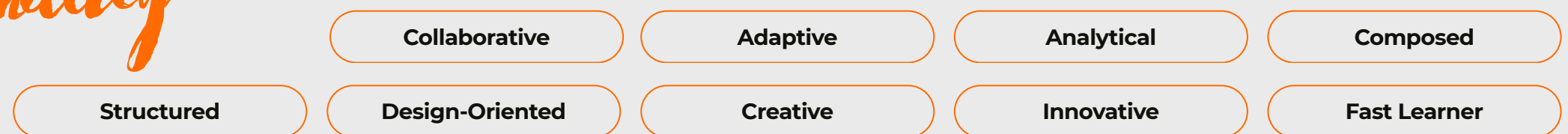
Software



Skills

- **Web Engineering & AI Integration:** Full-Stack Development (Laravel/PHP, Next.js, Tailwind CSS), Corporate Database Systems (MySQL), AI Automation (Gemini AI), Zero-Server-Load cPanel Deployment.
- **Interactive & WebAR Development:** Marker-based WebAR (MindAR, A-Frame), Virtual Tours (3DVista, WebXR/VR Mode), Educational Games, NPC AI (Behavior Tree Algorithm).
- **3D & Visual Asset:** 3D Modeling (Blender), 360° Image Stitching, Aerial Drone Videography, Environment Design, Motion Graphics.
- **UI/UX & Branding:** Prototyping (Figma), Frontend UX Optimization, Custom Interactive Audio Players, Visual Identity, Digital Marketing.
- **Production & Documentation:** Professional Video Editing, Asset Integration in Unity, Technical Reporting (Word), Data Management (Excel), Visual Presentation (PowerPoint).

Personality



Starting with a curiosity to bridge physical and digital worlds, my journey in Multimedia Engineering transforms complex theories into interactive realities. Through GenBI and PMM, I have evolved into a professional who balances structure and creativity to deliver technically sound, high-impact solutions.

introduction

Education

Politeknik Negeri Medan, Indonesia

Agustus 2021 - Oktober 2025

Majoring in Multimedia Graphics Engineering, Department of Computer and Informatics

GPA 3.81/4.00

Politeknik Negeri Malang, Indonesia

Agustus 2023 - Desember 2023

Teknik Informatika - Pertukaran Mahasiswa Merdeka (PMM) Batch 3

Engaged in cross-campus academic collaboration, enhancing adaptability, cross-cultural communication, and teamwork within a diverse academic environment.

Experiences

Revitalization of Barzanji Tradition & WebAR Navigation (Besilam Babussalam)

Lead WebAR Engineer & Multimedia Developer

Mei - Juli 2026

Engineered a marker-based WebAR wayfinding system (besilambabussalam.com) using MindAR and A-Frame for real-time navigation across 7 historical sites. Integrated Next.js, custom interactive audio players, and high-resolution 360° drone videography deployed on a zero-server-load cPanel architecture.

Virtual Tour & WebAR Integration (Virtual Maimun)

Multimedia & WebAR Developer

Mei - Juli 2026

Designed and deployed an interactive web-based virtual tour platform (virtualmaimun.com) using 3DVista Tour Builder. Integrated dynamic WebAR features to visualize interactive multimedia extensions for newly conserved heritage landmarks in Medan.

Stakeholder Database Management System (PT Pertamina Patra Niaga |

Project PIC & Full-Stack Engineer

Maret - Mei 2026

Engineered a corporate-scale stakeholder database and web application using Laravel and MySQL Cloud, integrating AI for automated data analysis and optimizing queries to eliminate server bottlenecks (Error 503)



Experiences

AR Wayfinding for Religious Tourism

Lead Developer & Researcher

Mei - Juli 2026

Developed a GPS & Marker-based navigation system using AR Foundation to enhance digital tourism.

Behavior Tree Research (Math Rescue)

NPC AI Researcher

Mei - Juli 2026

Optimized NPC logic using Behavior Trees (BT) to create more adaptive and realistic AI interactions.

GenBI (Bank Indonesia Scholarship)

Pubsosdok & ICT Division

Maret - Mei 2026

Managed digital assets and created numerous official certificates for national-scale events.

PT Ouzen Anugerah Indonesia

Digital Marketing Intern

Januari - Februari 2025

Developed aesthetic 3D motion graphics for product promotion, enhancing visual engagement.

Freelance Project Development

Multimedia Developer

2025

Created interactive AR Math applications and color recognition apps for early childhood education.

Rain Cycle Learning (AR for Deaf Students)

AR Developer

2024

Built an inclusive AR app that increased student learning comprehension by over 40%

"State of Sumatera" (Gemastik 2023)

2D Game Developer

2023

Collaborated in developing a cultural-themed side-scroller game for a national-level competition.

introduction



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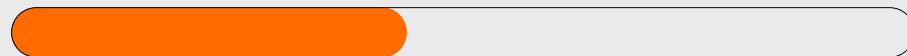
Languages

INDONESIAN



NATIVE

ENGLISH



TECHNICAL INTERMEDIATE

Lets Connect !!



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[Instagram.com/arz.dhana_](https://www.instagram.com/arz.dhana_)



github.com/Arzdhna007



arrizqi.rdhna@gmail.com



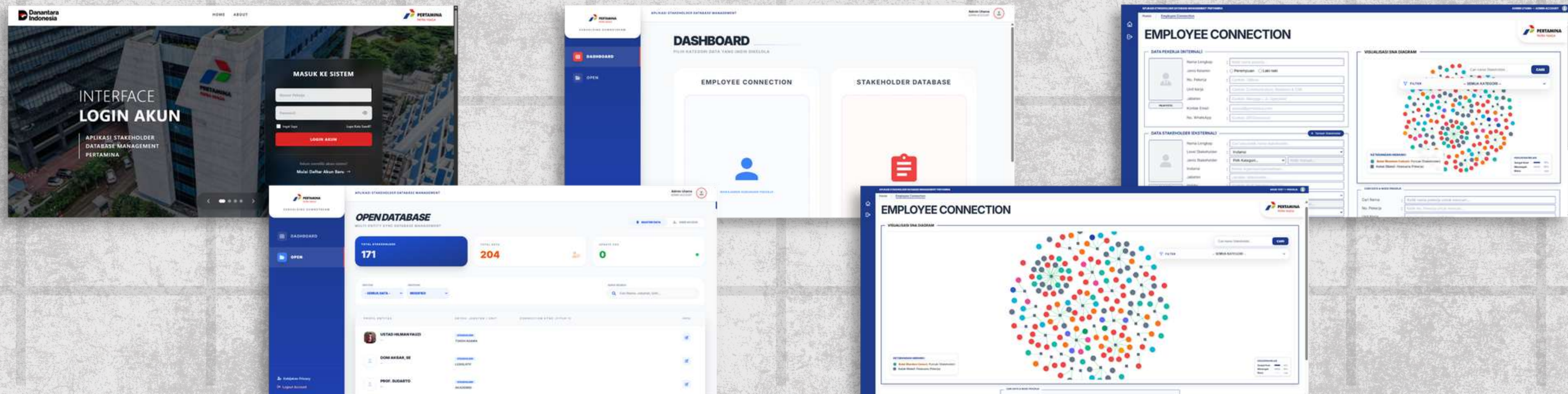
Stakeholder Database Management System

Project PIC & Full-Stack Engineer | ppn-stakeholdermanagement.com

Tools:     

In the Stakeholder Database Management System project for PT Pertamina Patra Niaga, I served as the Project PIC and Full-Stack Engineer, spearheading the end-to-end development of a corporate-scale web application (ppn-stakeholdermanagement.com). Built with Laravel (PHP), Tailwind CSS, and MySQL Cloud Hosting, the platform was engineered to streamline downstream stakeholder data management.

A core innovation of this project was integrating Gemini AI to automate data classification and analysis, significantly accelerating the development lifecycle. Furthermore, I executed complete server administration via cPanel managing custom domains and SMTP SSL authentication while optimizing database queries using Eager Loading and Pagination. This technical overhaul successfully eliminated server bottlenecks (Error 503), ensuring a highly stable, secure, and fully compliant live production environment under the Indonesian Data Protection Law (UU PDP No. 27/2022).



Barzanji Tradition Revitalization & WebAR Navigation

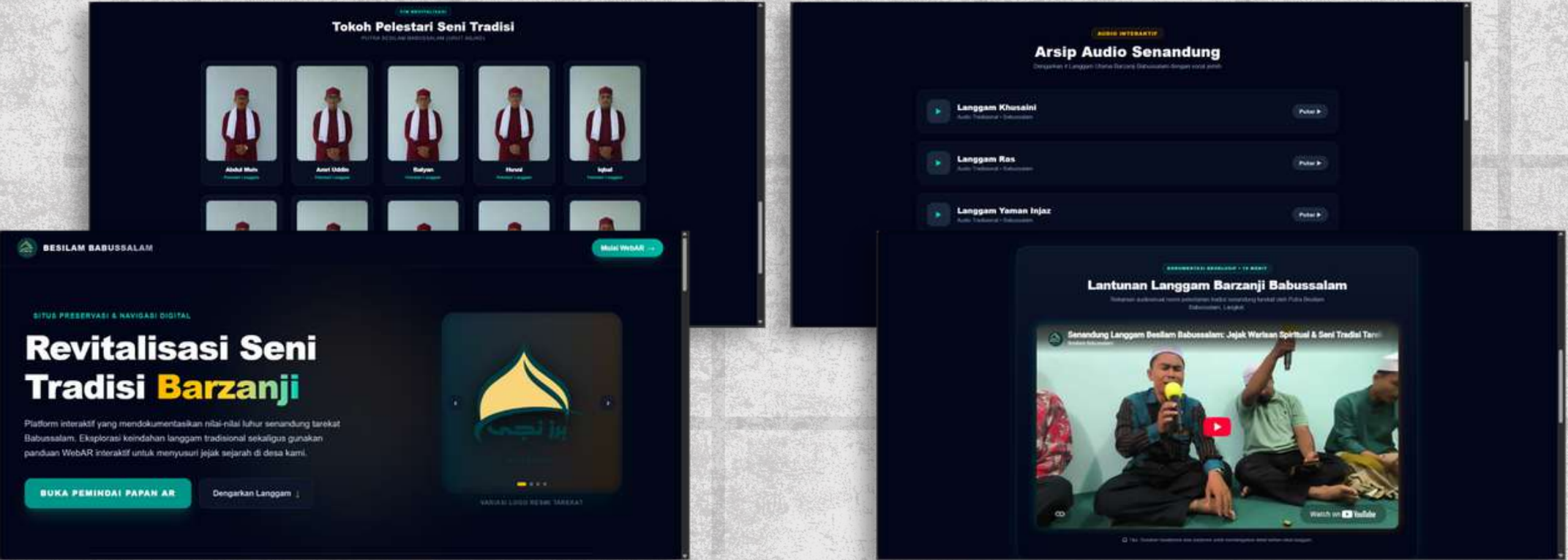
Lead WebAR Engineer & Multimedia Developer | besilambabussalam.com

Tools:

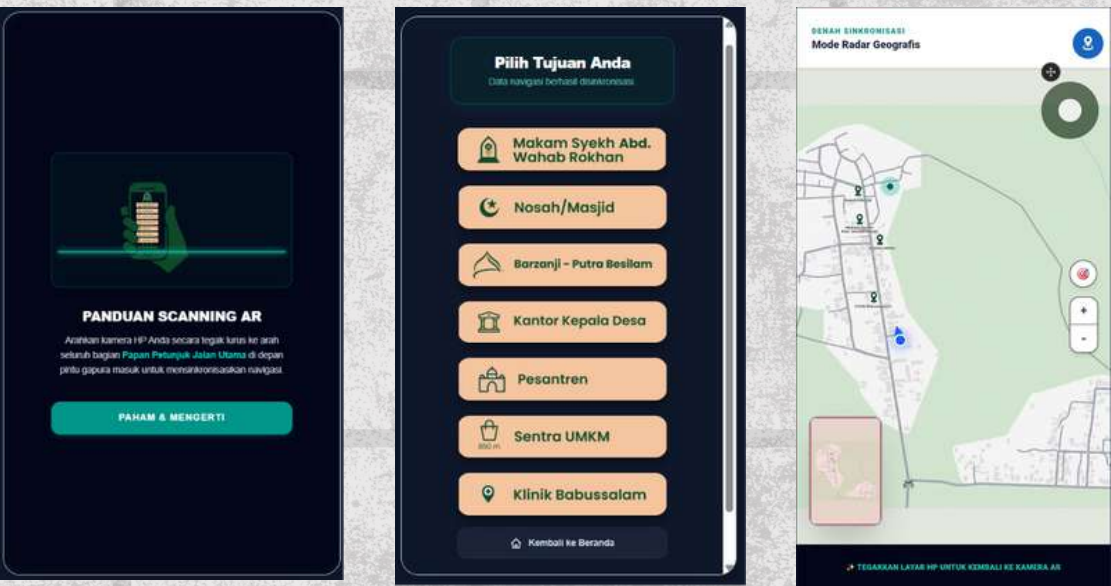


In this heritage revitalization project for Besilam Babussalam, I served as the Lead WebAR Engineer, developing an immersive real-time wayfinding system (besilambabussalam.com). Using MindAR and A-Frame, I engineered a marker-based navigation tool that scans physical signboards to accurately guide visitors across 7 historical landmarks.

I architected the educational portal using Next.js and Tailwind CSS, embedding privacy-enhanced videos and custom interactive audio players featuring asynchronous loading for traditional Barzanji recitations. To ensure peak UX and stability across mobile devices, I implemented dynamic hardware camera controls (front/rear toggle with a 400ms protection delay), LocalStorage state management, and executed a static export deployment to cPanel with a zero-server-load architecture.



Augmented Reality in Phone



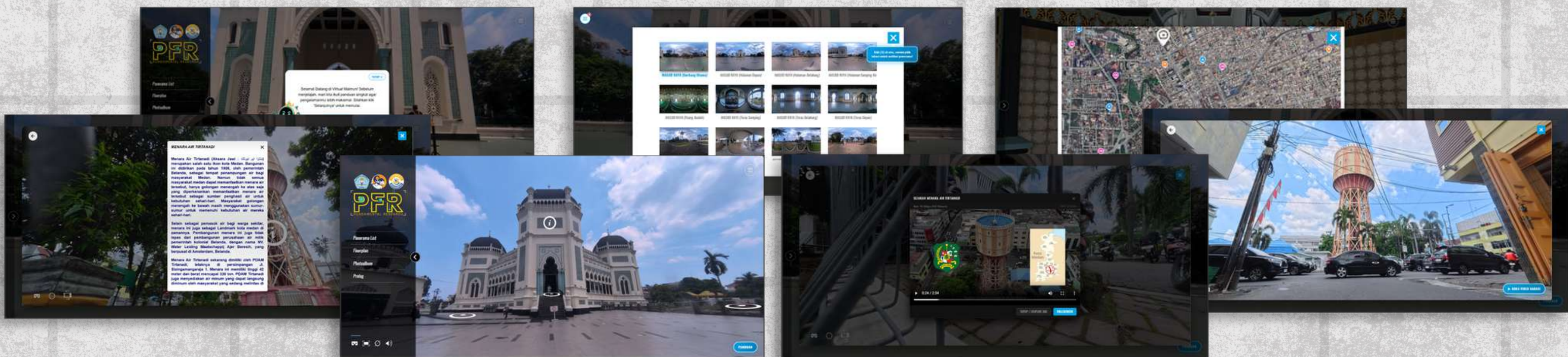
Virtual Tour & WebAR Integration

Multimedia & WebAR Developer | virtualmaimun.com

Tools:      

Spearheaded the multimedia and WebAR engineering for Virtual Maimun (virtualmaimun.com), an interactive web-based virtual tour platform designed to digitally preserve and showcase cultural heritage sites. Utilizing 3DVista Tour Builder, I structured a seamless, high-definition virtual exploration experience of the historic Maimun Palace environment, supported by independent 360° image stitching and aerial drone videography.

To expand the platform's digital footprint, I integrated dynamic WebAR extensions to visualize newly conserved surrounding landmarks including the Tirtanadi Water Tower, Juang Building, Ahmad Yani Park, and Pasar Merah. Furthermore, I optimized the frontend architecture by implementing custom iframe bypassers for hardware security restrictions and injecting tactical touch interceptors, guaranteeing flawless gyroscope sensors, VR mode (WebXR), and spotlight navigation across desktop and mobile devices.

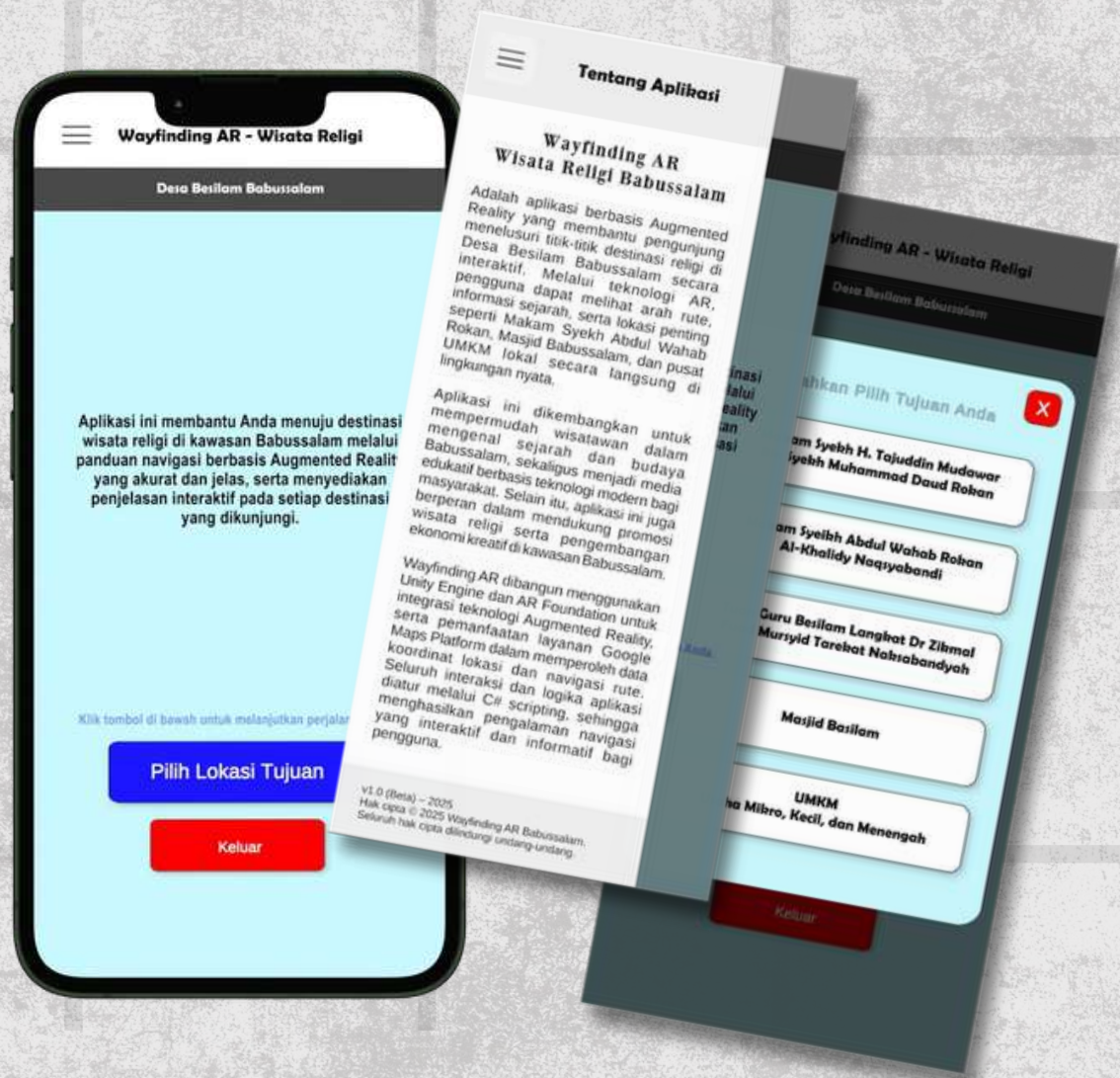


AR Wayfinding for Religious Tourism

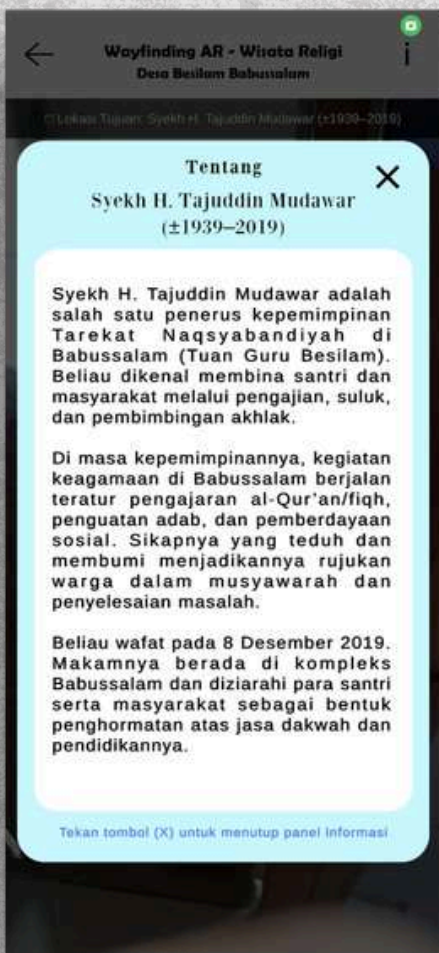
Lead Developer & Researcher

Tools:   

In the AR Wayfinding for Religious Tourism project, I spearheaded the development of an immersive navigation system using Unity and AR Foundation to revolutionize the visitor experience. By integrating a compass-based orientation and marker tracking system, I successfully transformed traditional directions into precise, real-time visual overlays that guide users interactively through complex sites. This project highlights my capability to bridge advanced technical engineering with user-centric design, ensuring that complex spatial data is delivered through a functional and educational digital tool.



Icon App



Behavior Tree Research (Math Rescue)

NPC AI Researcher

Tools:    

As the cornerstone of my Final Project (Undergraduate Thesis), I conducted an extensive technical analysis of Behavior Tree (BT) architectures to transcend the limitations of traditional Finite State Machines in RPG environments. My research focused on engineering high-fidelity NPC decision-making logic, enabling dynamic, goal-oriented responses that adapt fluidly to complex player interactions. By optimizing these AI frameworks, I successfully achieved a more sophisticated level of gameplay realism, proving my proficiency in advanced behavioral modeling to create deeply immersive and responsive educational experiences.



GenBI (Bank Indonesia Scholarship)

Pubsosdok & ICT Division

Tools:



As an awardee of the prestigious Bank Indonesia (GenBI) Scholarship, I served in the Pubsosdok & ICT Division, where I was entrusted with managing the community's digital identity and visual communication. My role involved leading the production of high-volume digital assets, including the design of numerous official certificates and documentation for national-scale programs. This experience honed my ability to deliver professional-grade visual content under strict organizational standards, ensuring that every technological initiative and creative output effectively supported the community's empowerment goals.



PT Ouzen Anugerah Indonesia

Digital Marketing Intern

During my professional internship at PT Ouzen Anugerah Indonesia, I specialized in Digital Marketing by bridging creative storytelling with high-end visual technology. My primary achievement involved engineering aesthetic 3D motion graphics and high-fidelity visual assets for premium product campaigns, specifically for the Colus Men skin-care line. By integrating sophisticated 3D modeling with seamless transitions and modern aesthetics, I transformed static product concepts into high-engagement video content that effectively captured the brand's essence and amplified audience reach.

3D Motion Elements:

COLUS MEN
+CARE



Tools:



Freelance Project Development

Multimedia Developer

Tools:    

As an Independent Multimedia Developer, I have engineered diverse educational tools, most notably an Interactive AR Math application and a Color Recognition app for early childhood development. My approach focuses on simplifying complex learning through intuitive UI/UX and immersive mechanics, ensuring that advanced technology remains accessible and engaging for young learners. These projects exemplify my ability to manage full-cycle development—from conceptual design to final deployment—consistently delivering functional, high-impact digital solutions independently.

Icons:



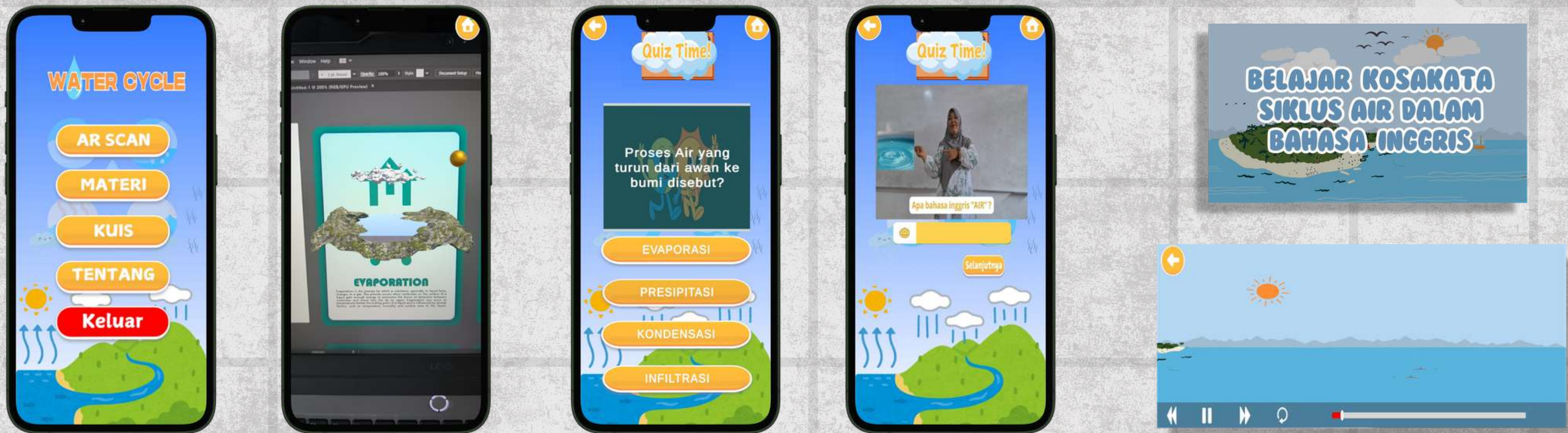
Rain Cycle Learning (AR for Deaf Students)

AR Developer

Tools:    

Driven by a commitment to inclusive education, I developed Rain Cycle AR, an augmented reality learning tool specifically engineered for deaf students. I focused on transforming auditory information into rich, interactive visual metaphors and sign-language-friendly interfaces to simplify complex environmental science concepts. This project demonstrates my expertise in user-centric design and my ability to leverage immersive technology to bridge communication gaps, ensuring that high-quality digital education is accessible to all learners regardless of their physical abilities.

learning materials:



Rain Cycle Learning (AR for Deaf Students)

AR Developer

Tools:    

Documentation:



“State of Sumatera” (Gemastik 2023)

2D Game Developer

Tools:   

In the 'State of Sumatera' project developed for the prestigious Gemastik 2023 national competition, I served as a 2D Game Developer dedicated to preserving Indonesian cultural heritage through interactive digital media. I was responsible for the end-to-end development of game mechanics and the creation of 2D visual assets, ensuring a seamless and engaging educational experience regarding Sumatran history. This project demonstrates my proficiency in balancing complex programming logic with artistic storytelling, delivering high-quality digital entertainment that meets competitive national standards.



Gameplay:

Tools:



Packaging Design



Game Logo



2D Character Sprite



Game UI Elements



App Logo Design



Asset Map Game Vector



Asset Background Game

Graphic Design 2D Multimedia Experience

In the realm of game development, I specialize in creating high-quality 2D Game Art and Assets that bring interactive stories to life. My work ranges from intricate character designs for the 'State of Sumatra' project to environment assets and UI elements tailored for immersive gameplay. I focus on maintaining a cohesive visual style and optimizing every asset for seamless integration within the Unity engine, ensuring that the artistic vision enhances the overall player experience.



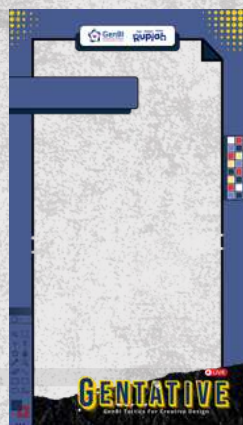
2D Character



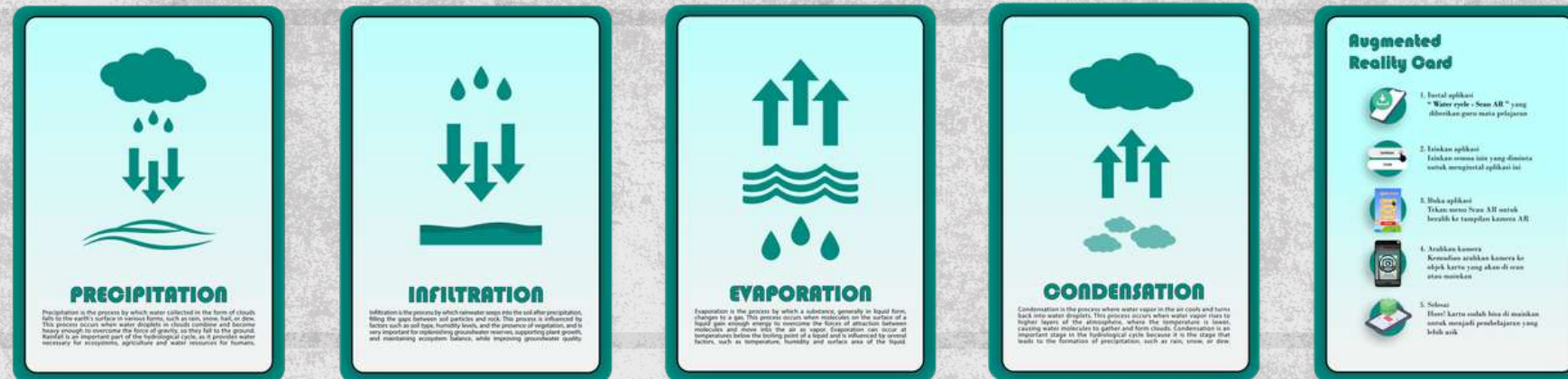
Design Poster Feed Instagram



Design Poster Film



Design Template Live Report



Design Smart Card AR



Design Poster

Parallel to my technical projects, I have extensive experience in Commercial Graphic Design, focusing on building strong brand identities and visual communications. This includes the development of Product Packaging for Lumpia Hot, cinematic Film Posters, and strategic Social Media content that effectively capture audience attention. By utilizing Adobe Illustrator and Photoshop, I combine modern aesthetics with marketing principles to deliver professional-grade visual assets that meet diverse industry standards.

THANK YOU

Arrizqi Pramadhana

Lets Connect !!



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